

Problem. On a cache hit, the executor must decide how much of the action head’s denoising schedule to skip. Let N be the number of denoising steps of the flow-matching action expert and let $\tau \in \{0, 1, \dots, N\}$ denote the skip depth: the integration is initialized at the retrieved intermediate state of depth τ and runs the remaining $N - \tau$ steps. In this view every hit is a warm start of some depth, and the familiar tiers are its extreme cases: full replay is the special case $\tau = N$, in which the retrieved chunk is executed directly, and full inference is the case $\tau = 0$, generation from pure noise. Convention: τ counts *skipped* steps; the executor’s start parameter is the remaining integration fraction, $\text{start_t} = (N - \tau)/N$. We fix this mapping here because sign errors between data generation, plots, and the executor are otherwise easy to make. The dispatch question is therefore how deeply to skip on each hit. Existing rules are coarse: our current dispatch quantizes τ to three values, while fixed-depth rules apply one global τ to every hit. We instead seek a per-step rule that allocates the remaining denoising computation without training a separate policy.

Two statistics. Both inputs are byproducts of the retrieval already performed. The first is the fused similarity s_t , which measures whether the current state is covered by the library. The second measures whether the retrieved neighborhood agrees on what to do. Let $a_{(1)}, \dots, a_{(k)}$ be the action chunks of the top- k retrieved entries, $\bar{a}$ their mean, and W a fixed diagonal (or Mahalanobis) weighting that standardizes the action dimensions; define the local action disagreement

$$v_t = \frac{1}{k} \sum_{j=1}^k \|W(a_{(j)} - \bar{a})\|^2. \quad (1)$$

Because the top- k entries come from neighboring states rather than repeated samples at the same state, v_t is not a pure measure of aleatoric uncertainty. It may reflect retrieval error, task-phase aliasing, action multimodality, teacher stochasticity, and library density. We therefore treat it as a local disagreement statistic whose predictive value is measured empirically.

Random loss under coupled noise. Let c denote the current conditioning (observation and prompt), let $x_{j,\tau}$ be the retrieved intermediate of depth τ from library entry j , and let z_j be the initial noise of the cached trajectory that produced it. Write $F_c^{\tau \rightarrow N}$ for the probability-flow integration from depth τ to completion under conditioning c . Define

$$a^{(\tau)} = F_c^{\tau \rightarrow N}(x_{j,\tau}), \quad a^{(0)} = F_c^{0 \rightarrow N}(z_j), \quad Y_\tau = \|W(a^{(\tau)} - a^{(0)})\|. \quad (2)$$

The reference $a^{(0)}$ is full generation under the *current* conditioning from the *same* initial noise z_j . The coupling matters because identical distribution does not mean identical realization: a freshly sampled $z' \sim \mathcal{N}(0, I)$ may land on a different action branch, and the measured deviation would then confound the warm-start mismatch with the teacher’s own sampling variability. By construction $Y_0 = 0$: at $\tau = 0$ the two branches coincide.

The calibration artifact therefore stores, for each cache entry (equivalently, each action-generation call), either the initial noise tensor or a deterministic seed. A 50×32 float32 tensor requires approximately 6.4 KB, which is small relative to the stored denoising intermediates. A seed is cheaper, but it requires bit-exact regeneration across the software and hardware configuration used for calibration. Current runtime payloads need not change: the additional field is required only while constructing the calibration data. As a preliminary diagnostic, we also measure the teacher’s noise-induced variability through repeated full generations at the same observation. If teacher–teacher deviation is negligible relative to warm-start deviation, an uncoupled analysis may be reported separately; its target is population deviation from a random teacher sample rather than the paired initialization error defined above.

Conditional quantiles. For $\tau \geq 1$ define

$$q_\tau(s, v) = Q_{1-\alpha}(Y_\tau \mid s_t = s, v_t = v), \quad (3)$$

the conditional quantile of the random loss given the retrieval statistics; it is a function of (τ, s, v) , not a quantile of an expectation. The qualitative motivation is empirical rather than theoretical: additional refinement steps may correct initialization mismatch, and a retrieved branch of a multimodal target can only be arbitrated by the steps that remain. We do not derive these effects from smoothness constants of the flow map; a Lipschitz bound alone does not imply contraction.

Why a threshold is not naive. Consider the binary decision “accept or reject a full replay” based on a fixed scalar score s , and define the offline-checkable event

$$Z = \mathbb{1}\{Y_N \leq \delta\}, \quad (4)$$

direct replay stays within the prescribed deviation tolerance. If the likelihood ratio of the conditional densities of s given $Z = 1$ and $Z = 0$ is monotone in s , then by the Neyman–Pearson lemma the threshold test on s is, at any fixed false-accept rate, optimal among all decision rules that depend on s alone; equivalently, thresholding traces the ROC frontier of the score. The statement is conditional on the fixed constraint and on the information available; it does not assert optimality against rules with access to more information. We use the deviation event because the effect of replaying one step on the eventual episode outcome is a closed-loop counterfactual that offline data cannot label. Assigning the episode outcome to every constituent step would not resolve this attribution problem. The surrogate is measurable on logged data, while task success is assessed through closed-loop evaluation. Improvements over the threshold must therefore come from one of three sources: additional statistics beyond s , better calibration of s , or a graded choice of τ instead of a binary one. The construction below addresses all three. The monotone-likelihood-ratio premise for this surrogate is assessed diagnostically on the fit split (E5).

Monotonicity assumption. For $\tau \geq 1$,

$$(A1) \quad q_\tau(s, v) \text{ is nondecreasing in } \tau \text{ and in } v, \text{ and nonincreasing in } s. \quad (5)$$

A sufficient and cleaner condition is first-order stochastic dominance of the conditional laws of Y_τ along each coordinate. (A1) is an assumption about the data-generating process, not a theorem; its empirical violation rate on the fit split is a reported diagnostic.

The dispatch surface. Fix a deviation tolerance δ and a confidence level $1 - \alpha$, and let $\tilde{q}_\tau(s, v)$, $\tau \geq 1$, denote the conformalized prediction bounds of Eq. 8. These bounds carry the episode-level marginal guarantee stated below; they are not pointwise conditional bounds at each (s, v) . Set $\tilde{q}_0 \equiv 0$, since $Y_0 = 0$ identically and full inference needs no statistical certificate. The rule is

$$\tau^*(s, v) = \max\left(\{0\} \cup \{\tau \geq 1 : \tilde{q}_\tau(s, v) \leq \delta\}\right). \quad (6)$$

The maximizing set always contains 0, so the rule is total; when the calibrated bounds exclude every positive depth, the step falls back to full inference. This is the fail-closed direction made explicit in the definition. Under (A1) the surface is bimonotone: nondecreasing in s , nonincreasing in v . The three coarse regimes are recovered as level sets rather than imposed: $\tau^* \rightarrow N$ where similarity is high and disagreement low, $\tau^* = 0$ where similarity is low, and intermediate τ^* where the state is covered but the neighborhood disagrees. The last region supplies a mechanistic reading of warm start that the three-tier rule lacked: partial denoising is retained exactly where the remaining steps have arbitration work to do.

Data separation and estimation. The experience library is built from a dedicated prefill set $\mathcal{D}_{\text{lib}}$ and frozen before dispatch calibration. Query episodes are then divided into three pairwise-disjoint sets—fit, calibration, and test—none of which contributes entries to the library. Same-trajectory retrieval is explicitly excluded. All divisions are made at the episode level because steps within an episode are strongly dependent.

The *fit* split is used to choose k , W , the depth grid, α , and δ ; fit the quantile surface; run the (A1) and monotone-likelihood-ratio diagnostics; and perform any pilot closed-loop selection. When δ is selected through pilot rollouts, an internal split or cross-validation within the fit data is used. The chosen operating point is frozen before final calibration. The *calibration* split is then used once to compute the conformal correction and is not revisited after any methodological change. The closed-loop *test* split is reserved for the final evaluation; a full frontier may be reported, but its primary operating point is declared in advance.

1. *Data.* For each step of the fit and calibration episodes, run the coupled reference once and each candidate depth $\tau \geq 1$ once, offline, recording $(s_i, v_i, Y_{i,\tau})$. This is replay of logged states, not interaction.

2. *Joint bimonotone fit.* On the fit split, estimate a single quantile surface $\hat{q}(\tau, s, v)$, $\tau \geq 1$, by minimizing the convex pinball (check) loss over the lattice-ordered monotone function class of (A1). Fitting each τ separately would allow the fitted surfaces to cross, producing depths at which a deeper skip appears safer than a shallower one; the joint constraint removes this artifact. Order-restricted convex optimization over a lattice partial order is solvable in polynomial time through linear-programming or network-flow formulations.
3. *Simultaneous conformal correction.* On the calibration split, define per-episode nonconformity scores that cover all steps and all positive depths at once,

$$R_e = \max_{i \in e} \max_{\tau \geq 1} (Y_{i,\tau} - \hat{q}(\tau, s_i, v_i)), \quad (7)$$

where i ranges over the steps of calibration episode e . Let c be the $\lceil (1 - \alpha)(|\mathcal{E}| + 1) \rceil$ -th order statistic of $\{R_e\}_{e \in \mathcal{E}}$ and set

$$\tilde{q}_\tau(s, v) = \hat{q}(\tau, s, v) + c, \quad \tau \geq 1. \quad (8)$$

The index must not exceed $|\mathcal{E}|$, which requires $|\mathcal{E}| \geq (1 - \alpha)/\alpha$ (at least 19 calibration episodes for $\alpha = 0.05$); otherwise $c = +\infty$ and the rule degenerates to $\tau^* \equiv 0$, which is valid but vacuous. This is a conformalized prediction bound with episode-level marginal coverage, rather than a conditional guarantee at a fixed (s, v) . Because the correction is simultaneous over depths and over the steps of an episode, the bound remains valid *after* the data-dependent selection of τ^* : under exchangeability of calibration and test episodes,

$$P\left(Y_{i,\tau} \leq \tilde{q}_\tau(s_i, v_i) \text{ for all steps } i \text{ and all } \tau \geq 1 \mid \text{a fresh episode}\right) \geq 1 - \alpha, \quad (9)$$

and in particular $Y_{i,\tau^*} \leq \delta$ holds jointly over the episode with probability at least $1 - \alpha$. Per-depth marginal calibration would not survive this selection step; the max construction is what makes the guarantee legitimate. The price is conservatism, which we report rather than hide. When several tasks are pooled, Eq. 9 is marginal coverage over the task mixture; per-task validity requires task-wise (Mondrian) calibration, and per-task results are reported where sample sizes permit.

Scope of the guarantee. Eq. 9 applies to fresh episodes drawn from the teacher visitation distribution represented by the calibration data. Deployment intervenes on the process: replayed prefixes alter subsequent visitation and may break exchangeability, reflecting the failure of composition described in the companion note. The guaranteed quantity is also an action-deviation proxy, not task success. We therefore study the association between δ and closed-loop failure risk on fit data, freeze the operating point before final calibration, and use closed-loop success on the test split as the final criterion. The conformal result provides an offline certificate under teacher visitation, not a guarantee for the intervened deployment distribution.

On the term “training-free”. The estimator is a nonparametric post-hoc calibration: isotonic quantile regression must minimize an objective over a class of monotone functions, so it is statistical fitting in the ordinary sense. What is absent is any policy or model fine-tuning, any gradient-based learner, and any parametric hypothesis class; the procedure is gradient-free, order-restricted convex optimization followed by quantile computation. We use “training-free” throughout in this restricted sense, consistent with the paper’s standing definition. A learned router (E10) remains in the study as the contrast: it asks whether gradient-based fitting buys anything beyond what calibrated order statistics provide.

Relation to existing rules. The rule family strictly contains the incumbents: the two-threshold dispatch is the degenerate surface that ignores v and quantizes τ to three values $\{0, \tau_{ws}, N\}$, and a fixed-depth rule is the constant surface $\tau \equiv \tau_0$ gated by a single threshold on s . Containment implies that the *oracle* optimum over surfaces is no worse than the oracle optimum over thresholds; it does not imply that a surface estimated from finite data beats a well-calibrated threshold, since the larger class can overfit. “Surface $\geq$ threshold” is therefore a hypothesis to be tested at matched calibration budgets, not an expected theorem. In the notation of the companion note, every surface policy is a causal scheduler over the retrieval information set, so its replay share remains a lower bound on $R_{\mathcal{C}(\mathcal{I}_\varphi)}(\varepsilon)$.

Experimental consequences. Evaluation proceeds in three stages. First, on one suite, we compare the surface restricted to the depths already stored online with a calibrated two-threshold rule under the same calibration budget. This pilot measures whether the episode-level simultaneous correction remains useful despite its conservatism: an overly large correction will route most states to $\tau^* = 0$ and eliminate any computational gain. Second, if the pilot shows a measurable advantage, we run the full comparison with the library, teacher, hardware, and calibration budget held fixed. The arms comprise a fixed threshold at fixed depth, the three-tier rule, the surface, the surface with stateful gating, a reduced-step teacher, and a learned dispatcher (E10). The primary readout is closed-loop success against measured GPU-time per decision step. Third, E5 reports the (A1) violation rate, the monotone-likelihood-ratio diagnostic for the surrogate event, and the association between deviation and closed-loop failure. All deviation data are obtained by offline replay of logged states together with the paired initial noise described above.